HIMANSHU NAIDU

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SKILLS

- **Programming Languages**: Python, JavaScript, Java, R, Swift, HTML5, C++
- **Python Frameworks**: TensorFlow, Keras, PyTorch, NumPy, Pandas, Scikit-learn, Geopandas, Pyspark
- **Databases**: PostgreSQL, MySQL, MongoDB, Firebase, NoSQL
- **Cloud Computing**: AWS EC2, AWS S3, AWS Neptune, Azure OpenAI Service, GCP
- **Other Tools and Libraries**: Spark, Git, Github, Docker, Jenkins, Kubernetes, TypeScript, Hadoop

EDUCATION

Master of Science, Data Science, University of Washington - Seattle, USA Sep 2023 – Apr 2025
Cumulative GPA: 3.98/4.00

Bachelor of Technology, Computer Science, VIT – Vellore, India Jun 2015 – Jun 2019
Cumulative GPA: 3.99/4.00

EXPERIENCE

Data Science Intern, eScience Institute - Seattle, USA Jun 2024 – Aug 2024

- Spearheaded the development of a comprehensive Python package to facilitate advanced predictive analytics of transit fare card (ORCA) data, enabling transit agencies to better understand transit usage patterns among diverse demographic groups.
- Developed heuristic and machine learning algorithms (e.g., ARIMA), along with in-depth Geospatial analysis, to identify areas in with gaps in low-income ridership, consequently guiding targeted distribution of low-income fare cards
- *Tools: PostgreSQL, PostGIS, Python, Statsmodels, GeoPandas, Shapely, Networkx, Tableau, Statistical Testing, Regression*

Graduate Research Assistant, Taskar Center for Accessible Technology - Seattle, USA Sep 2023 – Present

- Led a team of 5 developers to develop iOSPointMapper, an accessibility-focused application, leveraging GIS, LiDAR and Computer Vision, to enhance sidewalk mapping for individuals with disabilities.
- Implemented a Convolutional Neural Network (CNN) model in PyTorch, with a precision score (mAP) of 70, enabling on-device semantic segmentation for key accessibility points of interest.
- *Tools: Swift, PyTorch, Firebase, Convolutional Neural Network, Deep Learning, Agile Framework*

Software Engineer, ServiceNow - Hyderabad, India Dec 2021 – Jul 2023

- Optimized resource planning and allocation for over 3600 large-scale enterprises, through design and development of the Resource Management application using JavaScript and Web component technologies.
- Addressed over 40 customer support cases related to Resource Management and Timecard Management, leading to cost savings of over $2 million for customers (e.g., $110,000 for Medica), by fixing technical issues in tax reporting and timesheet cost calculations.
- *Tools: JavaScript, ServiceNow Platform, Seismic Framework, AngularJS, MariaDB, HTML, CSS*
- *Awards: Llama Best Performer Award, March 2023, for leading the Mobile App Development team*

Full-Stack Engineer, Societe Generale - Bengaluru, India Jan 2019 – Nov 2021

- Accomplished over 30% reduction in story delivery time for the Credit Risk team, facilitating better collaboration between business analysts and big data teams, by developing the MCG-HMI application on React.js and Node.js.
- Achieved a significant 71% average improvement in database response time by optimizing SQL procedures.
- Demonstrated the potential of Large Language Models, by presenting 6 Natural Language Processing (NLP) uses cases as a prompt engineer on OpenAI's GPT-3.
- *Tools: JavaScript, React, Node.js, PostgreSQL, MongoDB, HTML, CSS, Angular 7, GPT-3, Big Data Technologies*
- *Awards: Employee of Quarter, March 2021, for Full-Stack Development.*

PROJECTS

ClaimVer: Validation of Large Language Models Output through Evidence Attribution over Knowledge Graphs

- Fine-tuned LLMs to enhance claim verification over Knowledge Graphs, achieving a Rouge-L score of 0.694 and F1 score of 89.30.
- Enhanced the efficiency of Wikidata queries by deploying AWS Neptune endpoints on Wikidata dumps.

Identifying and Analyzing Crime Micro Hotspots in Seattle through Density Clustering

- Revealed smaller, high-density crime areas overlooked by traditional approaches by conducting clustering analysis using HDBSCAN and 5 other clustering algorithms to reveal.
- Analyzed the stability and evolution of the identified micro-clusters across census tracts by developing and applying an algorithm leveraging Wasserstein distance and Linear Sum Assignment.

Customer Churn Analysis

- Conducted a comprehensive analysis of customer churn in a fictionalized telecom company, identifying key drivers of churn through advanced statistical methods, including linear correlation checks, multicollinearity analysis, and logistic regression.
- Identified the most influential features affecting customer churn, by applying Lasso logistic regression, successfully narrowing down the feature set while maintaining high predictive accuracy (AUC score of 0.8637).